

🧠 Language Models

- Qwen3-Embedding-0.6B
- Qwen3-Embedding-4B
- Qwen3-Embedding-8B
- Google-Gemma-0.3B
- Intfloat-E5-Large-0.6B
- Intfloat-E5-Mistral-7B
- Nvidia-Llama-Nemotron-8B

🧠 UniversalCEFR Data

- ZAEBUC, Arabic, n=209
- Merlin, Czech, n=439
- CEFR-ASAG, English, n=299
- ICLE500, English, n=488
- ELLE, Estonian, n=1352
- Merlin, German, n=1033
- Merlin, Italian, n=806
- COPLE2, Portuguese, n=942
- PEAPL2, Portuguese, n=481

Embedding Probes

- Linear Regression
- Ordinal Regression
- MLP Regression
- Logistic Regression
- MLP Classifier

Validation Conditions

Identically Distributed

Validate performance on
20% hold-out data

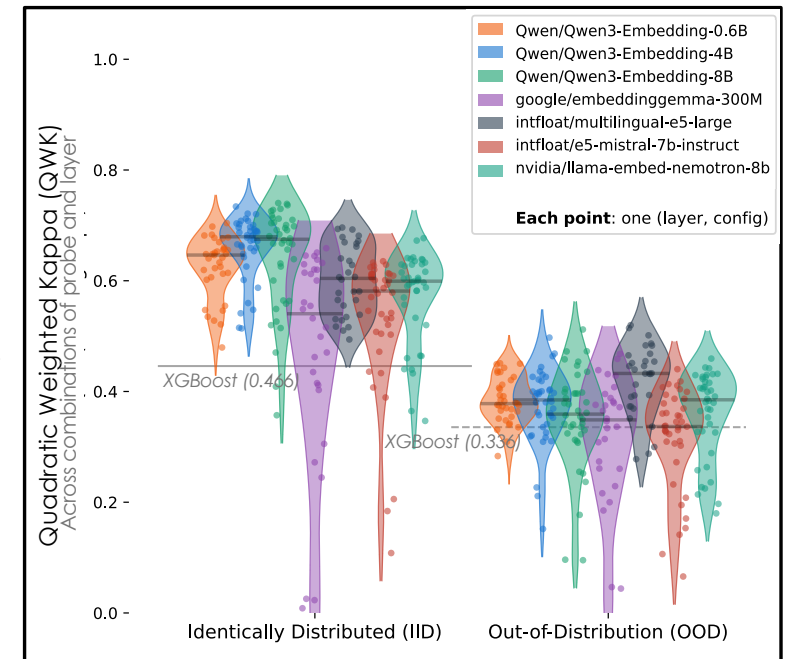
Out-of-Distribution

Validate performance on
leave-one-out dataset.

XGBoost

- Document Length
- Average Sentence Length
- Average Word Length

Significant Performance Drop



Violin plot showing performance distribution of each trained model.